

## Tools for calculating Multiple Sequence Alignments (Total of 252, 20.09.2023)

### Limitations of this overview

1. Collected are here only Publications about MSA algorithms, Programs and server I found during a simple text search for “multiple sequence alignment” on PubMed, resulting in over 15.000 Citations
2. There are probably publications that were missed due to missing keywords or different titles
3. I didn’t search for specific servers, e.g. The publications I found for MAFFT refer to the original algorithm and Version 5.0 aside others, so there are probably some publications I missed in between
4. I also didn’t read any Reviews about the topic to search for additional servers, since this simply would have taken too much time
5. Meta-Servers containing MSA Tools as part of their offerings were in most cases excluded from the search, even if they made use of new tools
6. Publications about improving parts of the algorithms, for example better Guide-Trees, that were then applied to existing methods were also excluded from the search
7. Normally you would need to look for additional literature about the following topics as well
  - a. Assessing MSAs
  - b. Analyzing MSAs as well as Tools for analyzing MSAs
  - c. Tools for visualizing MSAs like JalView, SeaView
  - d. Publications about the used Benchmarks
  - e. Publications about the theoretical background of the methods
  - f. Refining MSAs and Tools you could use for refining MSAs
  - g. Adjustments made to parts of the algorithms, e.g. different approaches for creating the Guide-Tree

### Short historical overview

- The first wave was from 1985 to 1995
- Around 1995 – 2005 reviews, databases, benchmarks, visualization tools, analysis tools, Meta-Server (PFAM, SMART) and assessment tools are developed
- The second wave was from 2000 to around 2008

- Starting around 2003/2004 algorithms for RNA/DNA and genomic sequences start to appear
- Starting around 2007/2008 less new tools appear instead you see adjustments made to already existing tools → The hype seems to be over
- Around 2013/2014 a third wave of programs (new and old) appeared, courtesy of a specialized issue of methods of molecular biology focused on MSA Tools
- With the growing success and availability of NGS data new tools for aligning nucleotide sequences emerged starting around 2018

### Role of MSAs

Multiple sequence alignments play a role in the following aspects of bioinformatics:

- Phylogenetic analysis
- Research of evolutionary history
- structure prediction of proteins and RNA molecules
- secondary structure assignment of protein structures
- function prediction of proteins
- domain / motif prediction of proteins
- development of secondary databases
- development of substitution matrices

### Tools without Publications

Clustal

GeneBee

Mix'n'Match

AMPS-Package

GCG

GenA1

MegALIGN

GeneJockeyII

Sequence Navigator

MUSCA

A(\*) Algorithm

PileUp

MGA

MSAID

EGMA

Pattern-constrained MSA

PSAlign

Camillo-and-Lipman-bounds

MUSTANG

MACAW

AlineaGA

KB-MSA

GramAlign

MSAGMOGA

BBOMP

MOMSA

MO-BFO

MASC

BDT-COFFEE

PSOSA

WITCH

DNPMSA

WMSA 2

## Tools with Publications

| Tool                | Paper1 | Paper2 | Paper3 | Paper4 | Paper5 | Paper6 | Paper7 | Paper8 |
|---------------------|--------|--------|--------|--------|--------|--------|--------|--------|
| 3dcoffee            | 2004   | 2004   |        |        |        |        |        |        |
| aba                 | 2004   |        |        |        |        |        |        |        |
| adsp                | 1992   |        |        |        |        |        |        |        |
| amap                | 2006   |        |        |        |        |        |        |        |
| asa                 | 1999   |        |        |        |        |        |        |        |
| affine-do           | 2012   |        |        |        |        |        |        |        |
| aliwaba             | 2006   |        |        |        |        |        |        |        |
| align-m             | 2004   |        |        |        |        |        |        |        |
| alignment-with-gaps | 1993   |        |        |        |        |        |        |        |
| antclustal          | 2003   |        |        |        |        |        |        |        |
| a§                  | 2005   |        |        |        |        |        |        |        |
| bclalign            | 2008   |        |        |        |        |        |        |        |
| beginnings          | 1989   | 1987   | 1989   | 1987   | 1989   | 1986   |        |        |
| blastalign          | 2004   |        |        |        |        |        |        |        |
| blockmsa            | 2007   |        |        |        |        |        |        |        |
| carna               | 2012   |        |        |        |        |        |        |        |
| chaos-dialign       | 2004   |        |        |        |        |        |        |        |
| chaos               | 2003   |        |        |        |        |        |        |        |
| clustalv            | 1992   |        |        |        |        |        |        |        |
| cmsa                | 2017   | 2002   | 2003   |        |        |        |        |        |
| coach               | 2004   |        |        |        |        |        |        |        |
| cobalt              | 2007   |        |        |        |        |        |        |        |
| coffee              | 1998   |        |        |        |        |        |        |        |
| cactus              | 2011   |        |        |        |        |        |        |        |
| centroidalign       | 2013   |        |        |        |        |        |        |        |
| cloud-coffee        | 2010   |        |        |        |        |        |        |        |

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|-----------------------------------|------|------|------|------|------|------|------|--|
| <b>clustalo</b>                   | 2011 | 2018 |      |      |      |      |      |  |
| <b>clustalw</b>                   | 2014 | 2004 | 2007 | 2003 | 2005 | 2005 | 1994 |  |
| <b>clustalx</b>                   | 1997 |      |      |      |      |      |      |  |
| <b>comalign</b>                   | 1999 |      |      |      |      |      |      |  |
| <b>consensus-matrix</b>           | 1992 |      |      |      |      |      |      |  |
| <b>constraints</b>                | 1996 |      |      |      |      |      |      |  |
| <b>contextual-alignment-model</b> | 2005 |      |      |      |      |      |      |  |
| <b>convexmsa</b>                  | 2016 |      |      |      |      |      |      |  |
| <b>dca</b>                        | 2000 | 1997 | 1998 |      |      |      |      |  |
| <b>dcalign</b>                    | 2023 |      |      |      |      |      |      |  |
| <b>dbclustal</b>                  | 2000 |      |      |      |      |      |      |  |
| <b>dead-end</b>                   | 1999 |      |      |      |      |      |      |  |
| <b>dialign-p</b>                  | 2004 |      |      |      |      |      |      |  |
| <b>dialign-tx</b>                 | 2008 | 2010 |      |      |      |      |      |  |
| <b>dialign-t</b>                  | 2005 |      |      |      |      |      |      |  |
| <b>dialign</b>                    | 2013 | 2004 | 2006 | 1999 | 1996 |      |      |  |
| <b>divide-and-conquer</b>         | 1995 |      |      |      |      |      |      |  |
| <b>edna</b>                       | 2013 |      |      |      |      |      |      |  |
| <b>erdd</b>                       | 2017 |      |      |      |      |      |      |  |
| <b>eulerian-path</b>              | 2005 |      |      |      |      |      |      |  |
| <b>expresso</b>                   | 2006 |      |      |      |      |      |      |  |
| <b>fame</b>                       | 2020 |      |      |      |      |      |      |  |
| <b>famsa</b>                      | 2016 |      |      |      |      |      |      |  |
| <b>fmalign</b>                    | 2004 |      |      |      |      |      |      |  |
| <b>fsa</b>                        | 2009 |      |      |      |      |      |      |  |
| <b>fuzzy-hmm</b>                  | 2006 |      |      |      |      |      |      |  |
| <b>gap4</b>                       | 2007 |      |      |      |      |      |      |  |
| <b>gmt</b>                        | 1999 |      |      |      |      |      |      |  |
| <b>gobics</b>                     | 2004 | 2005 |      |      |      |      |      |  |

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|--------------------------|------|------|------|------|--|--|--|--|
| graspaline               | 2013 |      |      |      |  |  |  |  |
| genalign                 | 1988 |      |      |      |  |  |  |  |
| genetic-algorithm        | 2015 | 2002 | 2016 | 1997 |  |  |  |  |
| gibbs-sampling           | 1993 |      |      |      |  |  |  |  |
| halign-ii                | 2017 |      |      |      |  |  |  |  |
| halign3                  | 2022 |      |      |      |  |  |  |  |
| halign                   | 2015 |      |      |      |  |  |  |  |
| hhmer                    | 2008 | 2009 | 2011 |      |  |  |  |  |
| hhsearch                 | 2005 |      |      |      |  |  |  |  |
| hmm-kalign               | 2007 |      |      |      |  |  |  |  |
| hmm                      | 2003 |      |      |      |  |  |  |  |
| hmmerge                  | 2023 |      |      |      |  |  |  |  |
| hsa                      | 2006 |      |      |      |  |  |  |  |
| handel                   | 2001 |      |      |      |  |  |  |  |
| hidden-markov-models     | 1994 |      |      |      |  |  |  |  |
| ibbomsa                  | 2016 |      |      |      |  |  |  |  |
| ilp                      | 2002 |      |      |      |  |  |  |  |
| ispalign                 | 2008 |      |      |      |  |  |  |  |
| insertions               | 2005 |      |      |      |  |  |  |  |
| kissa                    | 2009 |      |      |      |  |  |  |  |
| kmad                     | 2016 |      |      |      |  |  |  |  |
| kalign2                  | 2009 |      |      |      |  |  |  |  |
| kalign3                  | 2020 |      |      |      |  |  |  |  |
| kalignp-shu-elifsson-201 | 2011 |      |      |      |  |  |  |  |
| kalign                   | 2005 | 2006 |      |      |  |  |  |  |
| lagan                    | 2003 |      |      |      |  |  |  |  |
| m2align                  | 2017 |      |      |      |  |  |  |  |
| macse                    | 2011 |      |      |      |  |  |  |  |
| mafft-dash               | 2019 |      |      |      |  |  |  |  |

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|-------------------|------|------|------|------|------|------|------|------|
| <b>mafft</b>      | 2019 | 2013 | 2008 | 2010 | 2002 | 2005 | 2005 | 2018 |
| <b>magnolia</b>   | 2008 |      |      |      |      |      |      |      |
| <b>magos</b>      | 2006 |      |      |      |      |      |      |      |
| <b>magus</b>      | 2022 | 2021 | 2021 |      |      |      |      |      |
| <b>maligned</b>   | 1992 |      |      |      |      |      |      |      |
| <b>malign</b>     | 1996 | 1986 |      |      |      |      |      |      |
| <b>mango</b>      | 2007 |      |      |      |      |      |      |      |
| <b>map2</b>       | 2005 |      |      |      |      |      |      |      |
| <b>marna</b>      | 2005 |      |      |      |      |      |      |      |
| <b>mascot</b>     | 1993 |      |      |      |      |      |      |      |
| <b>mash</b>       | 1991 |      |      |      |      |      |      |      |
| <b>mass</b>       | 2003 |      |      |      |      |      |      |      |
| <b>mastr</b>      | 2007 |      |      |      |      |      |      |      |
| <b>match-box</b>  | 1992 | 1997 |      |      |      |      |      |      |
| <b>mavid</b>      | 2003 | 2004 | 2007 |      |      |      |      |      |
| <b>mcoffee</b>    | 2007 | 2006 |      |      |      |      |      |      |
| <b>meme</b>       | 1994 |      |      |      |      |      |      |      |
| <b>mishima</b>    | 2010 |      |      |      |      |      |      |      |
| <b>msa-ga</b>     | 2007 |      |      |      |      |      |      |      |
| <b>msa-pad</b>    | 2015 |      |      |      |      |      |      |      |
| <b>msacsa</b>     | 2008 |      |      |      |      |      |      |      |
| <b>msacompro</b>  | 2011 |      |      |      |      |      |      |      |
| <b>msaindelfr</b> | 2015 |      |      |      |      |      |      |      |
| <b>msam</b>       | 2009 |      |      |      |      |      |      |      |
| <b>msaprobs</b>   | 2016 | 2010 |      |      |      |      |      |      |
| <b>msarc</b>      | 2014 |      |      |      |      |      |      |      |
| <b>msasa</b>      | 1994 |      |      |      |      |      |      |      |
| <b>msat</b>       | 2004 |      |      |      |      |      |      |      |
| <b>msa</b>        | 1999 |      |      |      |      |      |      |      |

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| mtrap                         | 2010 |      |  |  |  |  |  |  |
| mulan                         | 2007 | 2005 |  |  |  |  |  |  |
| multan                        | 1989 |      |  |  |  |  |  |  |
| multicomp                     | 1992 |      |  |  |  |  |  |  |
| mummals                       | 2006 |      |  |  |  |  |  |  |
| muscle                        | 2004 | 2004 |  |  |  |  |  |  |
| musicme                       | 2005 |      |  |  |  |  |  |  |
| mxscarna                      | 2008 |      |  |  |  |  |  |  |
| mauve                         | 2004 |      |  |  |  |  |  |  |
| mergealign                    | 2012 |      |  |  |  |  |  |  |
| mirage                        | 2018 |      |  |  |  |  |  |  |
| model                         | 2004 |      |  |  |  |  |  |  |
| music                         | 2004 |      |  |  |  |  |  |  |
| mugsy                         | 2011 |      |  |  |  |  |  |  |
| mulblast                      | 1996 |      |  |  |  |  |  |  |
| multiclustal                  | 1999 |      |  |  |  |  |  |  |
| multiipmaker                  | 2003 |      |  |  |  |  |  |  |
| murlet                        | 2007 |      |  |  |  |  |  |  |
| muscle5                       | 2022 |      |  |  |  |  |  |  |
| nast                          | 2006 |      |  |  |  |  |  |  |
| nralign                       | 2009 |      |  |  |  |  |  |  |
| opal                          | 2007 |      |  |  |  |  |  |  |
| palign                        | 2010 |      |  |  |  |  |  |  |
| pastaspark                    | 2017 |      |  |  |  |  |  |  |
| pasta                         | 2018 | 2015 |  |  |  |  |  |  |
| pcma-pej-sadreyev-grishin-200 | 2003 |      |  |  |  |  |  |  |
| pima                          | 1992 |      |  |  |  |  |  |  |
| poa                           | 2004 | 2002 |  |  |  |  |  |  |
| praline2                      | 2019 |      |  |  |  |  |  |  |



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|---------------------------|------|------|------|------|------|--|--|--|
| <b>pralinetm</b>          | 2008 |      |      |      |      |  |  |  |
| <b>praline</b>            | 2002 | 2002 | 2005 | 2005 | 2003 |  |  |  |
| <b>prank</b>              | 2008 | 2008 |      |      |      |  |  |  |
| <b>prime</b>              | 2006 |      |      |      |      |  |  |  |
| <b>promals3d</b>          | 2014 | 2008 | 2008 |      |      |  |  |  |
| <b>promals</b>            | 2007 | 2007 |      |      |      |  |  |  |
| <b>prrp</b>               | 1996 |      |      |      |      |  |  |  |
| <b>psic</b>               | 1999 |      |      |      |      |  |  |  |
| <b>psitmcoffee</b>        | 2016 |      |      |      |      |  |  |  |
| <b>pairwise</b>           | 1993 |      |      |      |      |  |  |  |
| <b>parallel-computers</b> | 1993 |      |      |      |      |  |  |  |
| <b>phylat</b>             | 2012 |      |      |      |      |  |  |  |
| <b>phylegeny-methods</b>  | 2005 |      |      |      |      |  |  |  |
| <b>picxaa</b>             | 2010 |      |      |      |      |  |  |  |
| <b>pipealign</b>          | 2003 |      |      |      |      |  |  |  |
| <b>pnpprobs</b>           | 2016 |      |      |      |      |  |  |  |
| <b>proda</b>              | 2006 |      |      |      |      |  |  |  |
| <b>prographmsa+tr</b>     | 2013 |      |      |      |      |  |  |  |
| <b>prographmsa</b>        | 2012 |      |      |      |      |  |  |  |
| <b>propip</b>             | 2021 |      |      |      |      |  |  |  |
| <b>probalign</b>          | 2007 | 2006 |      |      |      |  |  |  |
| <b>probcons</b>           | 2005 |      |      |      |      |  |  |  |
| <b>probpf</b>             | 2019 |      |      |      |      |  |  |  |
| <b>probe</b>              | 1997 |      |      |      |      |  |  |  |
| <b>qalign</b>             | 2003 |      |      |      |      |  |  |  |
| <b>qfast</b>              | 1998 |      |      |      |      |  |  |  |
| <b>qmalign</b>            | 2010 |      |      |      |      |  |  |  |
| <b>qoma</b>               | 2006 |      |      |      |      |  |  |  |
| <b>quickprobs2</b>        | 2017 |      |      |      |      |  |  |  |

|               |      |      |  |  |  |  |  |  |
|---------------|------|------|--|--|--|--|--|--|
| quickprobs    | 2014 |      |  |  |  |  |  |  |
| raga          | 1997 |      |  |  |  |  |  |  |
| rascal        | 2003 |      |  |  |  |  |  |  |
| ralign        | 2006 |      |  |  |  |  |  |  |
| rbt-ga        | 2009 |      |  |  |  |  |  |  |
| rbt-l         | 2009 |      |  |  |  |  |  |  |
| rcoffee       | 2008 | 2008 |  |  |  |  |  |  |
| re-music      | 2007 |      |  |  |  |  |  |  |
| rnamountalign | 2020 |      |  |  |  |  |  |  |
| revtrans      | 2003 |      |  |  |  |  |  |  |
| saga          | 1996 |      |  |  |  |  |  |  |
| salign        | 2012 | 2004 |  |  |  |  |  |  |
| saracoffee    | 2014 |      |  |  |  |  |  |  |
| satchmo-js    | 2010 |      |  |  |  |  |  |  |
| satchmo       | 2003 | 2003 |  |  |  |  |  |  |
| sate          | 2014 |      |  |  |  |  |  |  |
| scarna-lm     | 2009 |      |  |  |  |  |  |  |
| sigma-2       | 2010 |      |  |  |  |  |  |  |
| sigma         | 2006 |      |  |  |  |  |  |  |
| sma           | 2005 |      |  |  |  |  |  |  |
| somap         | 1991 |      |  |  |  |  |  |  |
| sov           | 2006 |      |  |  |  |  |  |  |
| spark-msna    | 2019 |      |  |  |  |  |  |  |
| spem          | 2005 |      |  |  |  |  |  |  |
| squint        | 2007 |      |  |  |  |  |  |  |
| stamp         | 1992 |      |  |  |  |  |  |  |
| stral         | 2006 |      |  |  |  |  |  |  |
| seq7          | 1997 |      |  |  |  |  |  |  |
| seqanmcoffee  | 2008 |      |  |  |  |  |  |  |

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|-----------------------------|------|------|------|--|--|--|--|--|
| <b>sequence-profiles</b>    | 1994 |      |      |  |  |  |  |  |
| <b>simulated-annealing</b>  | 2005 | 1993 |      |  |  |  |  |  |
| <b>tba</b>                  | 2004 |      |      |  |  |  |  |  |
| <b>tcoffee</b>              | 2011 | 2000 | 2003 |  |  |  |  |  |
| <b>tm-aligner</b>           | 2017 |      |      |  |  |  |  |  |
| <b>time-warping</b>         | 2017 |      |      |  |  |  |  |  |
| <b>translatorx</b>          | 2010 |      |      |  |  |  |  |  |
| <b>triplet-optimization</b> | 2004 |      |      |  |  |  |  |  |
| <b>vdga</b>                 | 2011 |      |      |  |  |  |  |  |
| <b>vista</b>                | 2004 |      |      |  |  |  |  |  |
| <b>witch-ng</b>             | 2023 |      |      |  |  |  |  |  |
| <b>weighted-sums</b>        | 1989 |      |      |  |  |  |  |  |
| <b>x-ins-i</b>              | 2008 |      |      |  |  |  |  |  |
| <b>xalign</b>               | 1994 |      |      |  |  |  |  |  |
| <b>aln3nn</b>               | 2007 |      |      |  |  |  |  |  |
| <b>codaln</b>               | 2005 |      |      |  |  |  |  |  |
| <b>i-adhore</b>             | 2011 |      |      |  |  |  |  |  |
| <b>pmmulti</b>              | 2004 |      |      |  |  |  |  |  |
| <b>transalign</b>           | 2005 |      |      |  |  |  |  |  |

**Histogram of Publication dates**